

# BIOS 735: Multiple Testing and Post-Selection Inference

November 17, 2025

## What Is Multiple Testing?

**Many scientific studies test more than one hypothesis.**

- ▶ Instead of asking a single question, modern datasets allow us to examine **dozens, hundreds, or even thousands** of potential associations.
- ▶ Examples include: many pollutants, many genes, many biomarkers, many clinical outcomes.
- ▶ Each hypothesis test has a chance of producing a *false positive*.

When we run many tests, those chances accumulate. Even if all null hypotheses are true, some tests will appear “significant” purely by chance.

## Why Multiple Testing Matters

**Key issue:** Running many tests increases the likelihood of reporting results that are not real.

- ▶ If you test 20 hypotheses at  $\alpha = 0.05$ , you expect **1 false positive** by chance.
- ▶ If you test 1,000 hypotheses, you expect **50 false positives**.
- ▶ These spurious results can mislead research, policy, and clinical decisions.

As datasets grow (e.g., electronic health records, exposome studies, genomics), the problem becomes unavoidable — and motivates the need for careful error control.

## Public Health Example: Screening Many Risk Factors

### **Scenario: Identifying risk factors for childhood asthma**

A large epidemiologic study collects data on hundreds of potential exposures:

- ▶ Indoor air pollutants (mold, dust, VOCs)
- ▶ Outdoor pollutants (PM<sub>2.5</sub>, NO<sub>2</sub>, ozone)
- ▶ Maternal factors (smoking, stress, nutrition)
- ▶ Early-life exposures (breastfeeding, infections, antibiotics)
- ▶ Neighborhood characteristics (traffic, green space, housing)

Researchers test each exposure to see whether it is associated with asthma. Even if **none** are true risk factors, some will appear significant simply because so many are tested.

## Public Health Example: Thousands of Potential Risk Factors

### **Scenario: Exposome-wide association study (EWAS)**

Modern public health datasets can include **thousands of exposures** measured for each participant:

- ▶ Hundreds of chemicals from biomonitoring (pesticides, metals, plastics)
- ▶ Dozens of dietary variables captured from 24-hour recalls
- ▶ Clinical biomarkers (lipids, hormones, inflammation markers)
- ▶ Genomic biomarkers

Researchers examine each exposure separately to identify which ones relate to an outcome such as obesity, diabetes, or preterm birth.

## Type I and II Errors

|                 |                                    | <b>Actual Situation “Truth”</b>                              |                                                              |
|-----------------|------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
|                 |                                    | <b>H<sub>0</sub> True</b>                                    | <b>H<sub>0</sub> False</b>                                   |
| <b>Decision</b> | <b>Do Not Reject H<sub>0</sub></b> | <b>Correct Decision</b><br>$1 - \alpha$                      | <b>Incorrect Decision</b><br><b>Type II Error</b><br>$\beta$ |
|                 | <b>Reject H<sub>0</sub></b>        | <b>Incorrect Decision</b><br><b>Type I Error</b><br>$\alpha$ | <b>Correct Decision</b><br>$1 - \beta$                       |

$$\alpha = \Pr(\text{Type I Error}) \quad \beta = \Pr(\text{Type II Error})$$

## Why Multiple Testing Matters

Genomics/Neuroimaging/Big Data = Lots of Data = Lots of Hypothesis Tests

- ▶ A typical microarray experiment might result in performing 100000 separate hypothesis tests.
- ▶ If we use a standard p-value cut-off of 0.05, we'd expect 5,000 genes to be deemed “significant” by chance.

## Why Multiple Testing Matters

- In general, if we perform  $m$  hypothesis tests, what is the probability of at least 1 false positive?

$$\Pr(\text{Making an error}) = \alpha$$

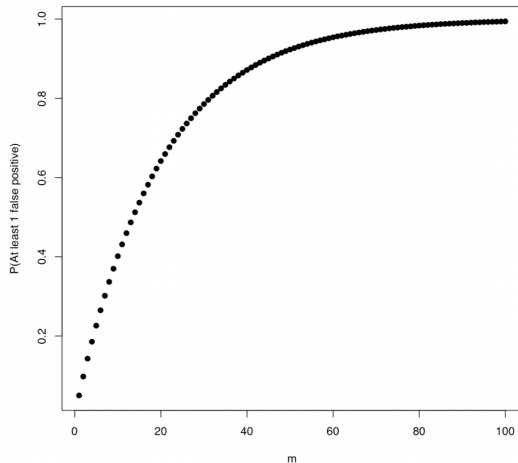
$$\Pr(\text{Not making an error}) = 1 - \alpha$$

$$\Pr(\text{Not making an error in } m \text{ tests}) = (1 - \alpha)^m$$

$$\Pr(\text{Making at least 1 error in } m \text{ tests}) = 1 - (1 - \alpha)^m$$



## Probability of At Least 1 False Positive



## Counting Errors

Assume we are testing  $H_1, H_2, \dots, H_M$ , where  $H_m = 1$  if the  $m$ th test is non-null.

|           | Not Rejected | Rejected | Total |
|-----------|--------------|----------|-------|
| $H_m = 0$ | $T$          | $V$      | $M_0$ |
| $H_m = 1$ | $U$          | $S$      | $M_1$ |
|           | $M - R$      | $R$      | $M$   |

- ▶  $M_0 = \#$  of True null hypotheses
- ▶  $M_1 = \#$  of False null hypotheses (Signals)
- ▶  $R = \#$  rejected hypotheses
- ▶  $V = \#$  of Type I errors (false positives)

## What Does Correcting for Multiple Testing Mean?

- ▶ When people say “adjusting p-values for the number of hypothesis tests performed” what they mean is controlling the Global Type I error rate of all hypotheses (in some fashion).
- ▶ There are many different methods in this area.
- ▶ Although these varied approaches have the same goal, they go about it in fundamentally different ways

## Different Approaches To Control Type I Errors

There are two main error rates that multiple testing procedures control:

- ▶ **Family-wise error rate (FWER)**: the probability of at least one type I error

$$FEWR = P(V \geq 1).$$

- ▶ **False discovery rate (FDR)**: the expected proportion of Type I errors among the rejected hypotheses

$$FDR = E(V/R | R > 0)P(R > 0).$$

## FWER

- ▶ Many procedures have been developed to control the Family Wise Error Rate (the probability of at least one type I error):
- ▶ Two general types of FWER corrections:
  1. **Single step:** equivalent adjustments made to each p-value
  2. **Sequential:** adaptive adjustment made to each p-value

## Single Step Approach: Bonferroni

- ▶ Very simple method for ensuring that the overall Type I error rate of  $\alpha$  is maintained when performing  $M$  independent hypothesis tests
- ▶ Rejects any hypothesis with p-value  $\leq \alpha/M$ .
- ▶ The adjusted p-values are:

$$\tilde{p}_j = \min(Mp_j, 1)$$

- ▶ For example, if we want to have  $FWER = 0.05$  when we perform 10,000 hypothesis tests, we'd need a p-value of  $0.05/10000 = 5 \times 10^{-6}$  to declare significance

## Objections to Bonferroni Corrections

*“Bonferroni adjustments are, at best, unnecessary and, at worst, deleterious to sound statistical inference”*

Perneger (1998)

- ▶ Counter-intuitive: interpretation of finding depends on the number of other tests performed
- ▶ The general null hypothesis (that all the null hypotheses are true) is rarely of interest
- ▶ High probability of type 2 errors, i.e. of not rejecting the general null hypothesis when important effects exist

Perneger, T. V. (1998). What's wrong with Bonferroni adjustments. *Bmj*, 316(7139), 1236-1238.

## FWER: Sequential Adjustments

- ▶ The simplest sequential method is Holm's Method
  - ▶ Order the unadjusted p-values such that  $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(M)}$
  - ▶ For control of the FWER at level  $\alpha$ , the step-down Holm adjusted p-values are

$$\tilde{p}_{(j)} = \min\{(M - j + 1)p_{(j)}, 1\}$$

- ▶ If  $\tilde{p}_{(j)} < \alpha$  then proceed to  $j + 1$ , otherwise stop.
  - ▶ Each  $\tilde{p}_{(j)}$  is multiplied by a different factor.
- ▶ For example, when  $M = 10000$ :

$$\tilde{p}_{(1)} = 10000p_{(1)}, \tilde{p}_{(2)} = 9999p_{(2)}, \dots, \tilde{p}_{(M-1)} = 2p_{(M-1)}, \tilde{p}_{(M)} = p_{(M)}$$



## Objections to FWER Corrections

- ▶ FWER is appropriate when you want to guard against ANY false positives
- ▶ However, in many cases (particularly in genomics/neuroimaging) we can live with a certain number of false positives
- ▶ In these cases, the more relevant quantity to control is the false discovery rate (FDR)

## Benjamini and Hochberg FDR

- ▶ To control FDR at level  $\alpha$ :
  - ▶ Order the unadjusted p-values such that  $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(M)}$
  - ▶ For a given  $\alpha$ , find the largest  $k$  such that  $p_{(k)} \leq \frac{k}{M}\alpha$ . That is, let

$$k = \arg \max_{i \in (1, \dots, M)} \left\{ i \cdot I \left( p_{(i)} \leq \frac{i\alpha}{M} \right) \right\} = \arg \max_{i \in (1, \dots, M)} \left\{ i \cdot I \left( \frac{M}{i} p_{(i)} \leq \alpha \right) \right\}.$$

- ▶ Reject the null hypothesis (i.e., declare discoveries) for all  $H_{(i)}$  for  $i = 1, \dots, k$ .

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- ▶ Reject the null hypothesis (i.e., declare discoveries) for all  $H_{(i)}$  for  $i = 1, \dots, k$ .
- ▶ The BH procedure is valid when the  $M$  tests are independent, and in some dependence scenarios (but not all). It has

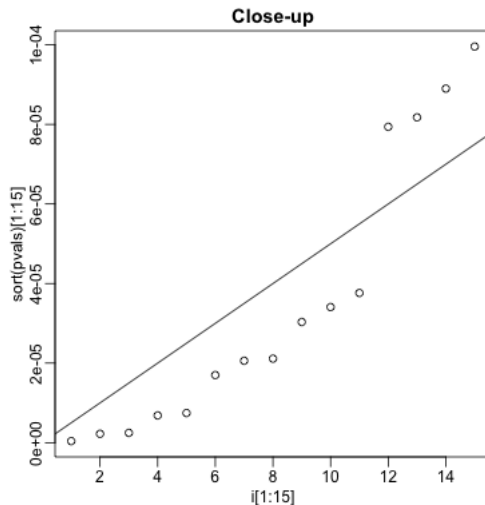
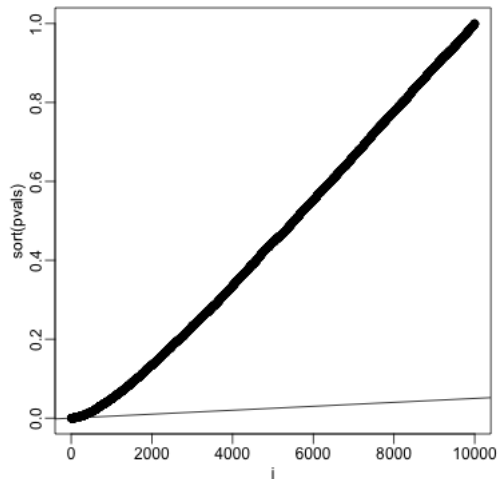
$$E(FDR_{BH}) \leq \alpha.$$

## BH FDR Example

| Rank (j) | $p_{(j)}$ | $\alpha \times \frac{j}{M}$ | $p_{(j)} < \alpha \times \frac{j}{M}$ | Reject $H_0$ ? | Adjusted p-value* |
|----------|-----------|-----------------------------|---------------------------------------|----------------|-------------------|
| 1        | 0.0008    | 0.005                       | Yes                                   | 1              | 0.008             |
| 2        | 0.009     | 0.01                        | Yes                                   | 1              | 0.045             |
| 3        | 0.016     | 0.015                       | No                                    | 1              | 0.048             |
| 4        | 0.019     | 0.02                        | Yes                                   | 1              | 0.048             |
| 5        | 0.196     | 0.025                       | No                                    | 0              | 0.392             |
| 6        | 0.35      | 0.03                        | No                                    | 0              | 0.583             |
| 7        | 0.541     | 0.035                       | No                                    | 0              | 0.772             |
| 8        | 0.781     | 0.04                        | No                                    | 0              | 0.976             |
| 9        | 0.9       | 0.045                       | No                                    | 0              | 0.989             |
| 10       | 0.993     | 0.05                        | No                                    | 0              | 0.993             |

\*the adjusted p-value is  $\min_{k \geq j} (Mp_{(k)}/k)$

## BH FDR Example



## Controlling for Multiple Testing in Practice

### How is multiple testing handled in real public health research?

- ▶ Different fields, journals, and study sections have **very different expectations**.
- ▶ Some require explicit corrections (e.g., Bonferroni, Holm, BH) whenever many tests are run, others allow unadjusted p-values if the analysis is exploratory or if results are framed cautiously.
- ▶ Large national studies and genomics/exposomics papers almost always expect corrections; smaller epidemiologic studies often do not.

**The reality:** Statistically, it's clear what one *should* do, but in practice there is a **gray area**—and expectations vary across reviewers, disciplines, and granting agencies.

**Takeaway:** You need to check what is typical or required for the specific journal, study section, or field you are writing for.

## Summary: Multiple Testing in a Nutshell

- ▶ Multiple testing adjustments control the global overall Type I error rate.
- ▶ Typically used when running many unadjusted analyses (i.e., t-tests between groups).
- ▶ **Key Error Rates:**
  - ▶ **FWER**,  $\Pr(\text{at least one false rejection})$ , Bonferroni, Holm.
  - ▶ **FDR**, proportion of type I errors, BH procedure.

### Takeaway

- ▶ Use **FWER control** when any false positive is costly.
- ▶ Use **FDR control** when testing many hypotheses and some false positives are acceptable.

## Post-Selection Inference: Movitation

- ▶ **Setting:** exploring the connection between dietary patterns and obesity with NHANES.
- ▶ **Scenario:** Researchers explore  $\sim 100$  dietary variables and use stepwise logistic regression to shrink them to 15. They report these predictors of obesity, with their p-values.
- ▶ Did they do anything wrong?



## Post-Selection Inference: Movitation

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- ▶ **Scenario:** Researchers explore  $\sim 100$  dietary variables and use stepwise logistic regression to shrink them to 15. They report these predictors of obesity, with their p-values.
- ▶ Did they do anything wrong?
- ▶ **Question:** how do you construct confidence intervals (CIs) or perform hypothesis testing when the data is used to select the variables and get the estimates/SEs?

## Post-selective inference

- ▶ Post-selective inference refers to the problem of making statistical inferences (like confidence intervals or hypothesis tests) after applying a model selection procedure.
- ▶ The challenge arises because traditional inferential procedures assume that the model structure (i.e., which predictors are included) is fixed in advance.
- ▶ The whole idea of model selection is to select the model structure based on the data.

## Post-selective inference

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- ▶ The challenge arises because traditional inferential procedures assume that the model structure (i.e., which predictors are included) is fixed in advance.
- ▶ The whole idea of model selection is to select the model structure based on the data.
- ▶ **Problems:**
  - ▶ Constructing CIs or performing hypothesis testing using standard methods after model selection leads to invalid inference (CI too narrow, increased type I errors).
  - ▶ Many procedures don't have "standard methods."

## Approaches

- ▶ **Invariance-based methods** for post-selective inference leverage the properties of certain statistics that remain invariant under model selection procedures.
- ▶ **Data Splitting** split the data into two parts. Use one part for model selection (e.g., determining which variables are non-zero with Lasso) and the other for inference.
- ▶ **Selective Inference Tools:** These methods condition the selection event (i.e., which variables were selected) when performing inference.
- ▶ **Conformal Inference:** a general tool for providing prediction intervals from any machine learning approach (any). It involves multiple refits of the model and can be computationally intensive.

## Data Splitting

- ▶ We'll discuss data splitting for the standard LASSO model:

$$Y = X\beta + \epsilon$$

where  $Y$  is the  $n \times 1$  response vector and

$$\hat{\beta} = \arg \min_{\beta \in R^p} \{ \|Y - X\beta\|_2^2 + \lambda \|\beta\|_1 \}$$

- ▶ Given a dataset of size  $n$ , randomly partition it into two subsets:
  - ▶ a **selection set** of size  $m$  and
  - ▶ an **inference set** of size  $n - m$ .

## Data Splitting with the LASSO

- ▶ Designate matrices and vectors:  $X_{sel}$ ,  $Y_{sel}$  for the selection set and  $X_{inf}$ ,  $Y_{inf}$  for the inference set.
- ▶ Fit a LASSO regression model to  $X_{sel}$  and  $Y_{sel}$ :

$$\hat{\beta}_{sel} = \arg \min_{\beta \in \mathbb{R}^p} \{ \|Y_{sel} - X_{sel}\beta\|_2^2 + \lambda \|\beta\|_1 \}$$

- ▶ The optimal  $\lambda$  value can be chosen using cross-validation within the selection set.
- ▶ Identify the set of predictors,  $S$ , where  $\hat{\beta}_{sel}$  has non-zero coefficients.

## Data Splitting with the LASSO

- ▶ Using the predictors in  $S$ , fit an ordinary least squares (OLS) regression model to  $X_{inf}$  and  $Y_{inf}$ .
- ▶ This will give unbiased estimates and valid statistical inferences, such as p-values and confidence intervals, since this dataset was not used in the model selection process.

## Data Splitting

- ▶ Data splitting is a simple method that yields valid inference because there is no “double dipping” into the data.
- ▶ However, because the data is split, the methods can suffer from reduced statistical power, especially if the dataset isn't large.
- ▶ Extensions:
  - ▶ consider multiple rounds of data splitting, the results from the different partitions can then be aggregated (e.g., by averaging) to obtain more robust estimates and inferences.
  - ▶ Advanced methods like cross-conformal prediction combine ideas from both data splitting and cross-validation.



## Selective Inference Tools

- ▶ Selective inference refers to the practice of making valid statistical inferences not just conditionally on the selected model but also considering the entire model selection process.
- ▶ These methods require complex mathematical derivations of the distribution of model quantities.
- ▶ We'll go through the basic idea for Lasso to give you the overall picture.

## Motivation for the Debiased Lasso

**Problem:** The standard Lasso is excellent for prediction and variable selection, but

- ▶ its coefficient estimates are **biased**, and
- ▶ classical inference (confidence intervals, p-values) is not valid.

**Why?** The  $\ell_1$  penalty shrinks coefficients toward zero, making the sampling distribution of  $\hat{\beta}_{\text{lasso}}$  difficult to characterize.

**Goal:** Construct an estimator that keeps the advantages of Lasso while enabling **asymptotically valid inference** in high-dimensional settings.

## Key Idea of the Debiased Lasso

**Debiased (or desparsified) Lasso:** Add a correction term to the Lasso estimator to remove the shrinkage bias.

$$\hat{\beta}^{debias} = \hat{\beta}^{lasso} + MX^{\top}(y - X\hat{\beta}^{lasso})/n$$

- ▶  $M$  is an approximate inverse of the Gram matrix  $(X^{\top}X/n)$ .
- ▶ The correction term approximates the *score function* from classical regression.

Result:  $\hat{\beta}^{debias}$  is approximately **unbiased and asymptotically normal**, allowing confidence intervals and hypothesis tests.

## What the Debiased Lasso Provides

### Assumptions (high-level)

- ▶ Approximate sparsity in the true coefficient vector
- ▶ Approximate sparsity in the inverse covariance structure
- ▶ Sub-Gaussian design (or similar tail conditions)

### Limitations

- ▶ Can be computationally heavy
- ▶ Inference is per-coordinate (not simultaneous)
- ▶ Performance degrades if the predictors are strongly correlated or non-sparse

## What Is Conformal Inference?

**Conformal inference** provides **distribution-free**, **finite-sample** prediction intervals for any regression or classification model.

- ▶ Works with **any** model: linear regression, random forest, neural nets, etc.
- ▶ Produces **valid coverage** under only an **i.i.d.** assumption.
- ▶ Gives intervals like  $[L(x), U(x)]$  such that

$$P(Y \in [L(x), U(x)]) \approx 1 - \alpha$$

**Goal:** Provide reliable uncertainty quantification without strong model assumptions.

## How Conformal Prediction Works

**Basic idea:** Use the model's own residuals to calibrate prediction intervals.

1. Fit a model on training data:  $\hat{f}(x)$ .
2. Compute absolute residuals on a calibration set:

$$r_i = |y_i - \hat{f}(x_i)|.$$

3. Take the  $(1 - \alpha)$  quantile of these residuals:

$$q = \text{Quantile}_{1-\alpha}(r_i).$$

4. Form prediction intervals:

$$[\hat{f}(x) - q, \hat{f}(x) + q].$$

**Result:** Guaranteed marginal coverage without assuming the model is correct.

## Why Conformal Inference Is Useful

### Advantages

- ▶ No distributional assumptions (beyond i.i.d.).
- ▶ Valid even for highly complex or black-box models.
- ▶ Easy to implement and computationally light.
- ▶ Works for regression, classification, and time-series variants.

### Limitations

- ▶ Intervals widen when model accuracy is poor.
- ▶ Standard conformal assumes i.i.d.; special methods needed for time dependence.
- ▶ Provides **marginal** (not conditional) coverage guarantees.